2ND SEMESTER LAB REPORT

TITLE : Electrical Circuit Analysis Lab



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BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

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	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 12

Lab Title: I

Objectiv: NORTON THEOREM

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 8/May/2025 Signature:

LABORATORY EXPERIMENT NO. 12

TO VERIFY MAXIMUM POWER TRANSFER THEOREM FOR DIFFERENT CONFIGURATION OF DC CIRCUITS

OBJECTIVE:

1: To check the validity of Maximum Power Transfer Theorem for Thévenin/Norton circuits.

2:To verify the results through software using Proteus.

Maximum Power Transfer Theorem

"A load will receive maximum power from a linear dc network when its total resistive value is exactly equal to the Thévenin resistance of the network as "seen" by the load."

OR:

"A load will receive maximum power from a linear dc network when its total resistive value is exactly equal to the Norton resistance of the network as "seen" by the load."

========(Lab Tasks)========

Lab task 1:

1. Analyse the circuit using Maximum Power Transfer Theorem:

a) Construct the Thevenin equivalent circuit of the previous lab, also shown below. (Do not fix the value of R L.) b) Take four resistors of different values (with one of the value equal to RL). Calculate the resistance of each resistor, using the Resistor Colour Codes. Record the values in four separate Data Tables and shown earlier in the solved example.

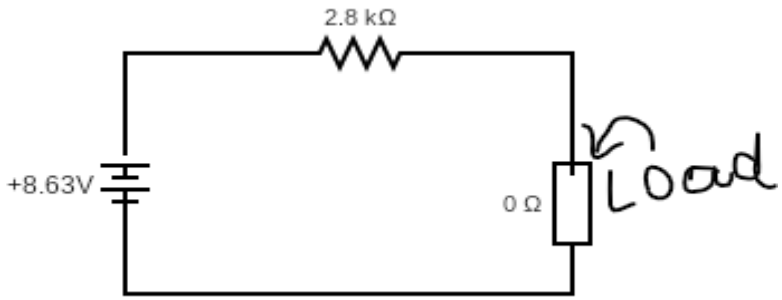
c) Now construct the circuit as shown at the right, taking one of the above mentioned resistors as R L.

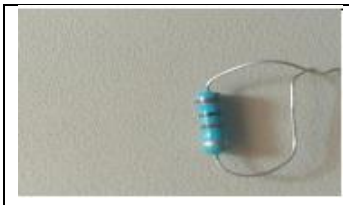
d) After verifying that he circuit is properly patched on the breadboard, switch on the trainer.

e) Measure the value of V L and I L using a DMM, calculate the value of P L. Record these values against the respective value of the resistor. (Show the theoretical calculations of the circuit, using all of the four different resistors one by one, as R L.)

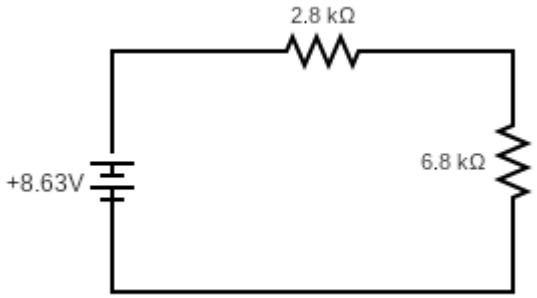
f) Repeat step (e) for the different resistors and record each V L , P L and I L against the respective R L .

g) Now turn off the power of the trainer and DMM.



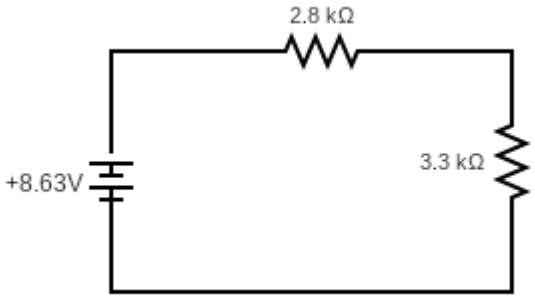
			
Resistor 1	**Resistor 2**	**Resistor 3**	**Resistor 4**
Blue: 6	**Orange: 3**	**Brown: 1**	**Red: 2**
Grey: 8	**Orange :3**	**Brown: 1**	**Grey:8**
Red: 2(100)	**Red: 2 (100)**	**Red: 2(100)**	**Red: 2(100)**
6800Ω (6.8KΩ)	3300Ω (3.3KΩ)	1100Ω (1.1kΩ)	2800Ω (2.8KΩ)

Thevenin Circuit 1



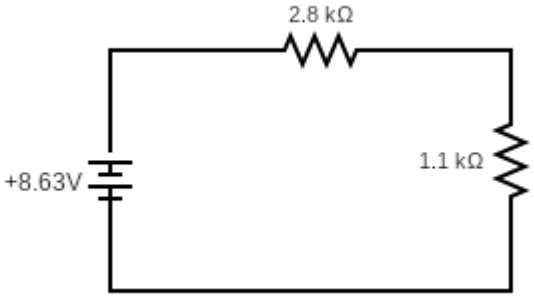
	Rth	RL	Total
Voltage(V)	2.52V	6.11	8.63
Current(mA)	0.90	0.90	0.90
Resistance(Ω)	2.8KΩ	6.8KΩ	9.6KΩ
Power(Watt)	2.268mW	5.499mW	7.767mW

Thevenin Circuit 2



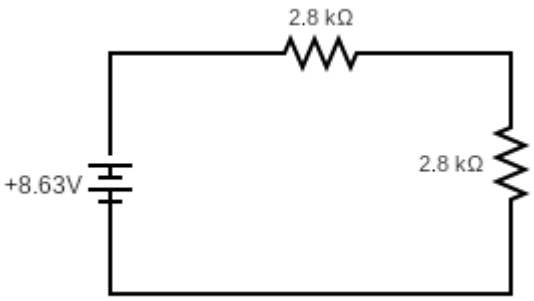
	Rth	RL	Total
Voltage(V)	3.96V	4.67V	8.63
Current(mA)	1.41	1.41	1.41
Resistance(Ω)	2.8KΩ	3.3KΩ	6.1KΩ
Power(Watt)	5.5836mW	6.58mW	12.16mW

Thevenin Circuit 3



	Rth	RL	Total
Voltage(V)	6.20V	2.43V	8.63V
Current(mA)	2.21	2.21	2.21
Resistance(Ω)	2.8KΩ	1.1KΩ	3.9KΩ
Power(Watt)	13.702mW	5.37mW	19.07mW

Thevenin Circuit 4



	Rth	RL	Total
Voltage(V)	4.31V	4.31V	8.63V
Current(mA)	1.54	1.54	1.54
Resistance(Ω)	2.8KΩ	2.8KΩ	5.6KΩ
Power(Watt)	6.637mW	6.637mW	13.27mW

Theoretical Calculation

Thevenin Circuit 1: (6.8KΩ)	**Thevenin Circuit 2: (3.3KΩ)**
Given: Rth = 2.8 kΩ = 2800 Ω Voc = Vth = 8.63 V RL = 6.8 kΩ = 6800 Ω **Step 1: Calculate total resistance** Rtotal = Rth + RL = 2800 Ω + 6800 Ω = 9600 Ω **Step 2: Calculate current through the circuit (Ith = IL)** IL = Vth / Rtotal = 8.63 V / 9600 Ω = 0.000899 A = 0.899 mA **Step 3: Calculate voltage across RL** VL = IL × RL = 0.000899 A × 6800 Ω = 6.113 V **Step 4: Calculate voltage across Rth** VRth = IL × Rth = 0.000899 A × 2800 Ω = 2.517 V **Step 5: Verify Kirchhoff's voltage law** Vth = VRth + VL 8.63 V = 2.517 V + 6.113 V 8.63 V = 8.63 V ✓ **Step 6: Calculate power delivered to RL** Power1 = VL × IL = 6.113 V × 0.000899 A = 0.00550 W = 5.50 mW Step 7: Calculate power delivered by Thévenin source Power2 = Vth × Ith = 8.63 V × 0.000899 A = 0.00776 W = 7.76 mW	Given: Rth = 2.8 kΩ = 2800 Ω Voc = Vth = 8.63 V RL = 3.3 kΩ = 3300 Ω **Step 1:** Rtotal = Rth + RL = 2800 Ω + 3300 Ω = 6100 Ω **Step 2:** IL = Ith = Vth / Rtotal = 8.63 V / 6100 Ω = 0.001415 A = 1.415 mA **Step 3:** VL = IL × RL = 0.001415 A × 3300 Ω = 4.67 V **Step 4:** VRth = IL × Rth = 0.001415 A × 2800 Ω = 3.96 V **Step 5:** Vth = VRth + VL = 3.96 V + 4.67 V = 8.63 V ✓ **Step 6:** Power1 = VL × IL = 4.67 V × 0.001415 A = 0.00661 W = 6.61 mW **Step 7:** Power2 = Vth × Ith = 8.63 V × 0.001415 A = 0.01221 W = 12.21 mW

Thevenin Circuit 3: (1.1KΩ)	Thevenin Circuit 4: (2.8KΩ)
Given: • Rth = 2.8 kΩ • Vth = 8.63 V • RL = 1.1 kΩ **Step 1**: Calculate IL using voltage divider: IL = Vth/(Rth + RL) = 8.63/(2800 + 1100) = 8.63/3900 = 0.002213 A = 2.213 mA **Step 2:** Calculate VL: VL = IL × RL = 0.002213 × 1100 = 2.434 V **Step 3:** Calculate ITh: ITh = IL = 2.213 mA **Step 4:** Calculate Power1: Power1 = VL × IL = 2.434 × 0.002213 = 0.00538 W = 5.38 mW **Step 5:** Calculate Power2: Power2 = Vth × ITh = 8.63 × 0.002213 = 0.0191 W = 19.1 mW	Given: • Rth = 2.8 kΩ • Vth = 8.63 V • RL = 2.8 kΩ **Step 1:** Calculate IL using voltage divider: IL = Vth/(Rth + RL) = 8.63/(2800 + 2800) = 8.63/5600 = 0.00154 A = 1.54 mA **Step 2**: Calculate VL: VL = IL × RL = 0.00154 × 2800 = 4.315 V **Step 3:** Calculate ITh: ITh = IL = 1.54 mA **Step 4:** Calculate Power1: Power1 = VL × IL = 4.315 × 0.00154 = 0.00665 W = 6.65 mW **Step 5:** Calculate Power2: Power2 = Vth × ITh = 8.63 × 0.00154 = 0.01329 W = 13.29 mW

Maximum Power Transfer Theorem Analysis

Given Thevenin Equivalent:

- Rth = 2.8 kΩ = 2800 Ω
- Vth = 8.63 V

Maximum Power Transfer Condition:

RL = Rth = 2.8 kΩ for maximum power transfer

Circuit 1: RL = 6.8 kΩ **Calculations:** • Rtotal = Rth + RL = 2800 + 6800 = 9600 Ω • IL = Vth/Rtotal = 8.63/9600 = 0.000899 A • VL = IL × RL = 0.000899 × 6800 = 6.113 V • PL = VL × IL = 6.113 × 0.000899 = **5.50 mW** **Efficiency:** η = PL/Psource = PL/(Vth × IL) = 5.50/(8.63 × 0.899) = **70.9%**	**Circuit 2: RL = 3.3 kΩ** **Calculations:** • Rtotal = Rth + RL = 2800 + 3300 = 6100 Ω • IL = Vth/Rtotal = 8.63/6100 = 0.001415 A • VL = IL × RL = 0.001415 × 3300 = 4.67 V • PL = VL × IL = 4.67 × 0.001415 = **6.61 mW** **Efficiency:** η = PL/Psource = 6.61/(8.63 × 1.415) = **54.1%**
Circuit 3: RL = 1.1 kΩ **Calculations:** • Rtotal = Rth + RL = 2800 + 1100 = 3900 Ω • IL = Vth/Rtotal = 8.63/3900 = 0.002213 A • VL = IL × RL = 0.002213 × 1100 = 2.434 V • PL = VL × IL = 2.434 × 0.002213 = **5.38 mW** **Efficiency:** η = PL/Psource = 5.38/(8.63 × 2.213) = **28.2%**	**Circuit 4: RL = 2.8 kΩ (MAXIMUM POWER CONDITION)** **Calculations:** • Rtotal = Rth + RL = 2800 + 2800 = 5600 Ω • IL = Vth/Rtotal = 8.63/5600 = 0.00154 A • VL = IL × RL = 0.00154 × 2800 = 4.315 V • PL = VL × IL = 4.315 × 0.00154 = **6.65 mW** **Efficiency:** η = PL/Psource = 6.65/(8.63 × 1.54) = **50.0%**

Maximum Power Transfer Verification:

Theoretical Maximum Power:

$P_{L,max} = V_{th}^2/(4 \times R_{th}) = (8.63)^2/(4 \times 2800) = 74.477/11200 =$ **6.65 mW** ✓

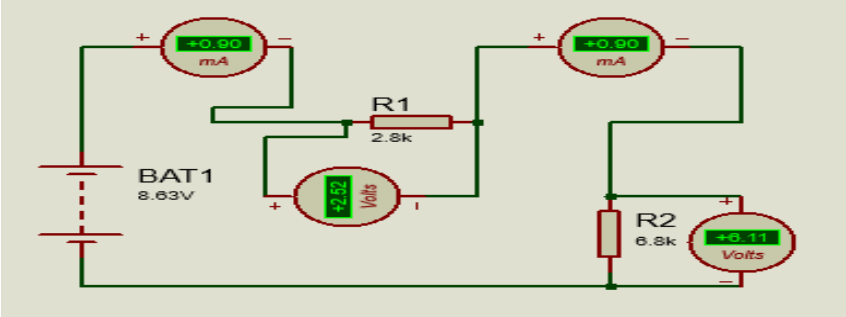
Summary Table:

Circuit	RL (kΩ)	Current (mA)	VL (V)	Power (mW)	Efficiency (%)
1	6.8	0.899	6.113	5.50	70.9
2	3.3	1.415	4.67	6.61	54.1
3	1.1	2.213	2.434	5.38	28.2
4	2.8	1.54	4.315	6.65	50.0

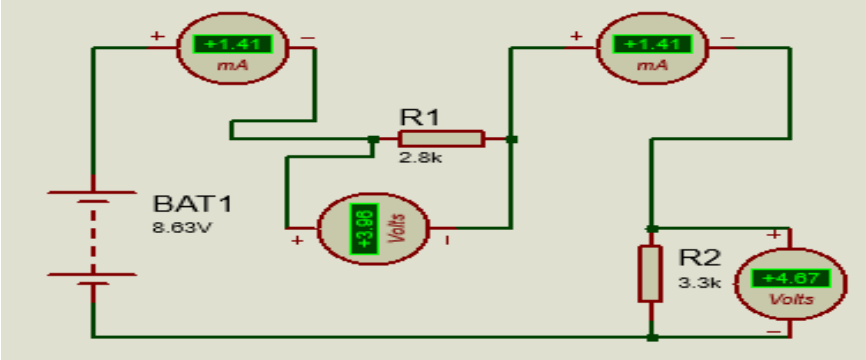
- **Maximum power transfer occurs when RL = Rth = 2.8 kΩ**
- **Maximum power delivered = 6.65 mW**
- **Efficiency at maximum power transfer = 50%**
- Circuit 4 delivers maximum power to the load
- Circuit 1 has highest efficiency but lower power transfer

Protius Simulation

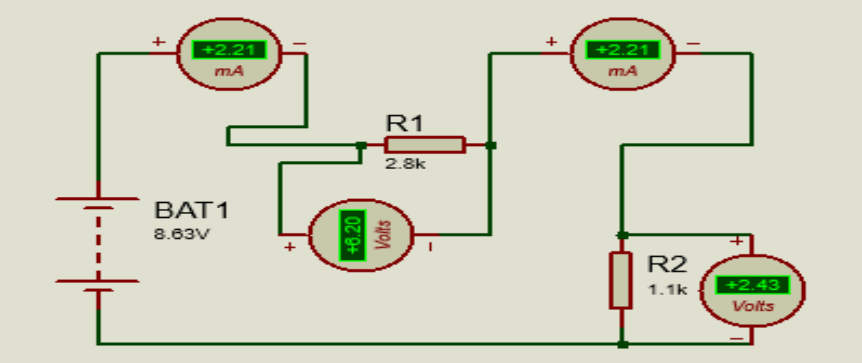
Thevenin Circuit 1: (6.8KΩ)



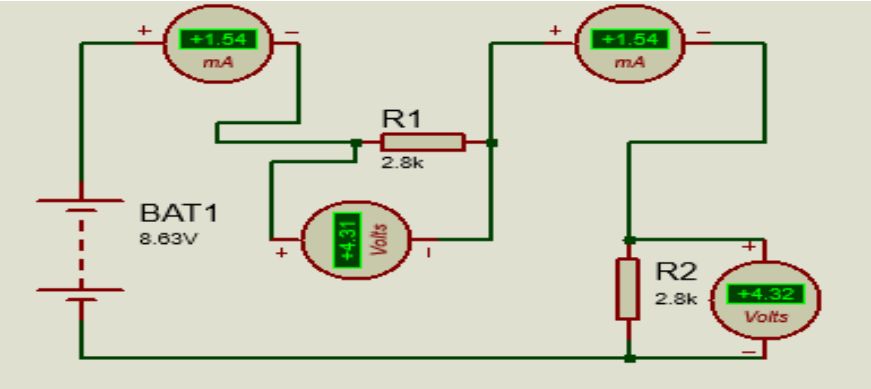
Thevenin Circuit 2(3.3KΩ)



Thevenin Circuit 3: (1.1KΩ)



Thevenin Circuit 4: (2.8KΩ)

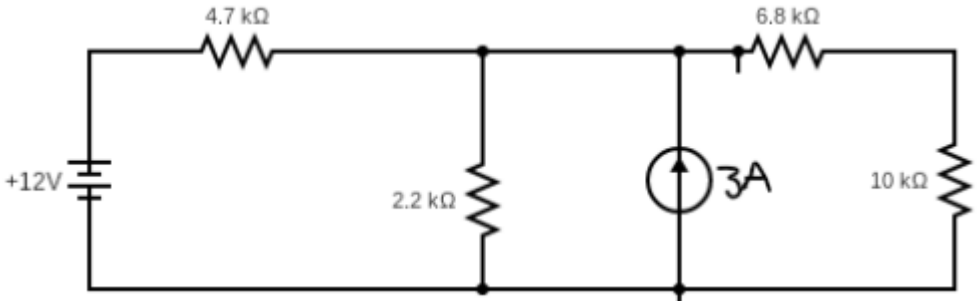


Lab Task 2:

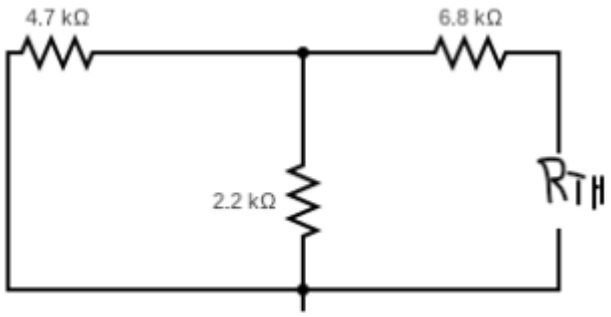
1. Analyse the circuit using Maximum Power Transfer Theorem:

For the circuit from Lab 10, Task 2, do the maximum power transfer theorem analysis on the Thevenin Equivalent circuit, Take four resistor values (one being equal to R L). Draw the thevenin equivalent circuit diagram, the detailed procedural steps as mentioned in the previous task, the data tables, theoretical calculations and Proteus Simulations.

CIRCUIT DIAGRAM



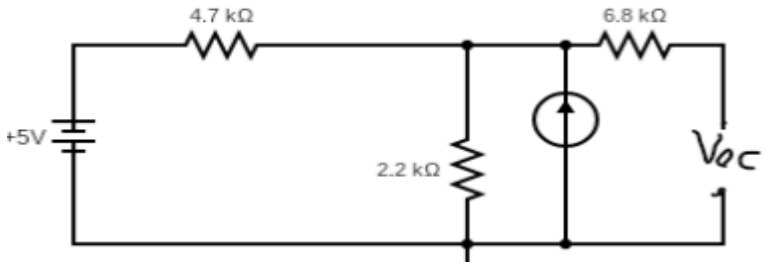
The portion of the circuit outlined is to be replaced by it Thévenin equivalent circuit.
To determine R th, set all sources to zero by replacing voltage source by short-circuit and current source by an open circuit, as shown:



= (2I1

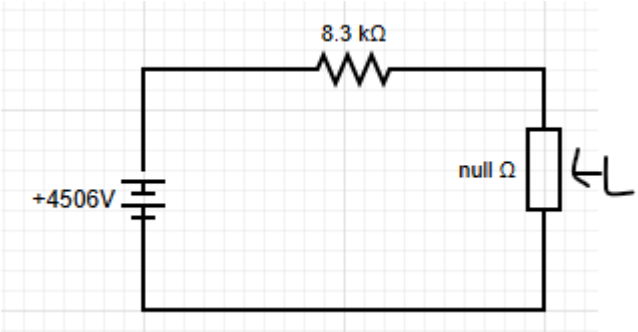
Rth=(4700‖2200)+6800

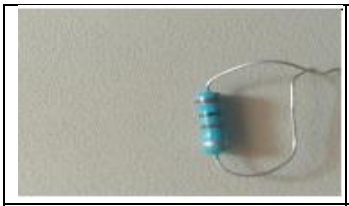
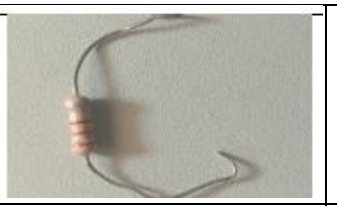
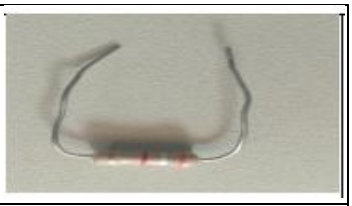
Rth ≈ 8298.55Ω



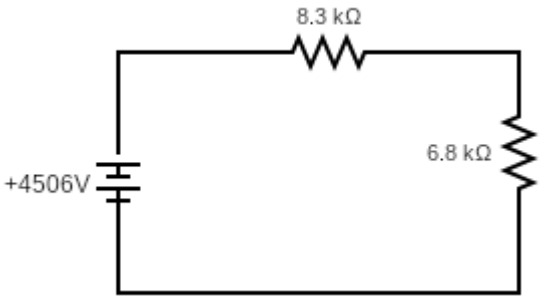
$$(v - 12) / 4700 + v / 2200 = 3$$

$$VTH \approx 4506$$



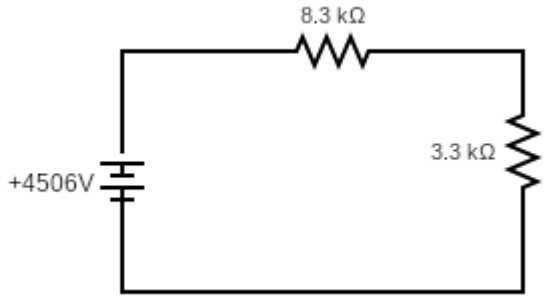
			
Resistor 1	**Resistor 2**	**Resistor 3**	**Resistor 4**
Blue: 6	**Orange: 3**	**Brown: 1**	**Grey: 8**
Grey: 8	**Orange :3**	**Brown: 1**	**Orange :3**
Red: 2(100)	**Red: 2 (100)**	**Red: 2(100)**	**Red: 2(100)**
6800Ω (6.8KΩ)	3300Ω (3.3KΩ)	1100Ω (1.1kΩ)	2800Ω (8.3KΩ)

Thevenin Circuit 1



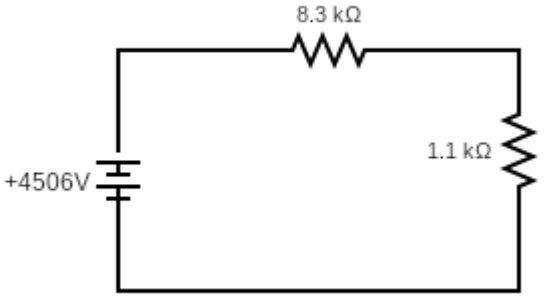
	Rth	**RL**	**Total**
Voltage(KV)	2.480KV	2.03KV	4.51KV
Current(A)	0.3A	0.3A	0.3A
Resistance(Ω)	8.3KΩ	6.8KΩ	15.1KΩ
Power(Watt)	744W	609W	1353W

Thevenin Circuit 2



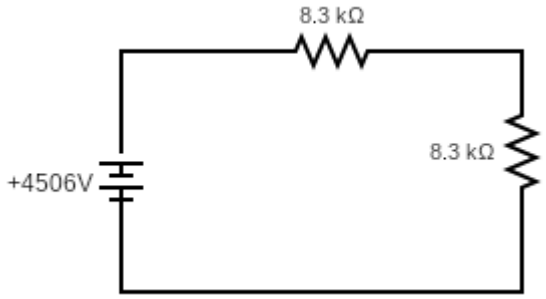
	Rth	RL	Total
Voltage(KV)	3.22KV	1.28KV	4.51KV
Current(A)	0.39A	0.39A	0.39A
Resistance(Ω)	8.3KΩ	3.3KΩ	11.6KΩ
Power(Watt)	1255.8W	499.2W	1755W

Thevenin Circuit 3



	Rth	RL	Total
Voltage(KV)	3.98KV	0.53KV	4.51KV
Current(A)	0.48A	0.48A	0.48A
Resistance(Ω)	8.3KΩ	1.1KΩ	9.4KΩ
Power(Watt)	1910W	254.4W	2164W

Thevenin Circuit 4



	Rth	RL	Total
Voltage(KV)	2.25KV	2.25KV	4.51KV
Current(A)	0.27A	0.27A	0.27A
Resistance(Ω)	8.3KΩ	8.3KΩ	16.6KΩ
Power(Watt)	607.5W	607.5W	1215W

➢ Maximum Power Transfer Theorem Check

- For maximum power transfer, RL should equal Rth

In this circuit, RL = 6.8 kΩ ≠ Rth = 8.3 kΩ

In this circuit, RL = 3.3 kΩ ≠ Rth = 8.3 kΩ

- In this circuit, RL = 1.1 kΩ ≠ Rth = 8.3 kΩ

- Therefore, the condition for maximum power transfer is not satisfied

➢ Maximum Power Calculation

Pmax = $(Vth^2) / (4 \times Rth) = (4506\ V)^2 / (4 \times 8300\ \Omega) = 20304036 / 33200 = 611.57\ W$

➢ Verify Power Calculation

Thevenin Circuit 1: (6.8KΩ)

Power1 = 605.5W < Pmax = 611.57 W ✓

Thevenin Circuit 2: (3.3KΩ)

Power1 = 497.5W < Pmax = 611.57 W ✓

Thevenin Circuit 3: (1.1KΩ)

Power1 = 252.7 W < Pmax = 611.57 W ✓

Thevenin Circuit 4: (8.3KΩ)

If - In this circuit, RL = 8.3kΩ = Rth = 8.3 kΩ

Power1 = 611.5 W =Pmax = 611.57 W ✓

Thevenin Circuit 1: (6.8KΩ)	**Thevenin Circuit 2: (3.3KΩ)**
Given: Rth = 8.3 kΩ = 8300 Ω Voc = Vth = 4.506 kV = 4506 V RL = 6.8 kΩ = 6800 Ω **Step 1:** Rtotal = Rth + RL = 8300 Ω + 6800 Ω = 15100 Ω **Step 2:** IL = Ith = Vth / Rtotal = 4506 V / 15100 Ω = 0.2984 A = 298.4 mA **Step 3:** VL = IL × RL = 0.2984 A × 6800 Ω = 2029.12 V **Step 4:** VRth = IL × Rth = 0.2984 A × 8300 Ω = 2476.88 V **Step 5:** Vth = VRth + VL = 2476.88 V + 2029.12 V = 4506 V ✓ **Step 6:** Power1 = VL × IL = 2029.12 V × 0.2984 A = 605.5 W **Step 7:** Power2 = Vth × Ith = 4506 V × 0.2984 A = 1344.6 W	Given: Rth = 8.3 kΩ = 8300 Ω Voc = Vth = 4.506 kV = 4506 V RL = 3.3 kΩ = 3300 Ω **Step 1**: Rtotal = Rth + RL = 8300 Ω + 3300 Ω = 11600 Ω **Step 2:** IL = Ith = Vth / Rtotal = 4506 V / 11600 Ω = 0.3884 A = 388.4 mA **Step 3:** VL = IL × RL = 0.3884 A × 3300 Ω = 1281.72 V **Step 4:** VRth = IL × Rth = 0.3884 A × 8300 Ω = 3224.28 V **Step 5:** Vth = VRth + VL = 3224.28 V + 1281.72 V = 4506 V ✓ **Step 6:** Power1 = VL × IL = 1281.72 V × 0.3884 A = 497.82 W **Step 7:** Power2 = Vth × Ith = 4506 V × 0.3884 A = 1750.13 W

Thevenin Circuit 3: (1.1KΩ)	**Thevenin Circuit 4: (8.3KΩ)**
Given: Rth = 8.3 kΩ = 8300 Ω Voc = Vth = 4.506 kV = 4506 V RL = 1.1 kΩ = 1100 Ω **Step 1:** Rtotal = Rth + RL = 8300 Ω + 1100 Ω = 9400 Ω **Step 2:** IL = Ith = Vth / Rtotal = 4506 V / 9400 Ω = 0.4793 A = 479.3 mA **Step 3:** VL = IL × RL = 0.4793 A × 1100 Ω = 527.23 V **Step 4:** VRth = IL × Rth = 0.4793 A × 8300 Ω = 3978.77 V **Step 5:** Vth = VRth + VL = 3978.77 V + 527.23 V = 4506 V ✓ **Step 6:** Power1 = VL × IL = 527.23 V × 0.4793 A = 252.7 W **Step 7:** Power2 = Vth × Ith = 4506 V × 0.4793 A = 2159.7 W	Given: Rth = 8.3 kΩ = 8300 Ω Voc = Vth = 4.506 kV = 4506 V RL = 8.3 kΩ = 8300 Ω **Step 1:** Rtotal = Rth + RL = 8300 Ω + 8300 Ω = 16600 Ω **Step 2:** IL = Ith = Vth / Rtotal = 4506 V / 16600 Ω = 0.2714 A = 271.4 mA **Step 3:** VL = IL × RL = 0.2714 A × 8300 Ω = 2253 V **Step 4:** VRth = IL × Rth = 0.2714 A × 8300 Ω = 2253 V **Step 5:** Vth = VRth + VL = 2253 V + 2253 V = 4506 V ✓ **Step 6:** Power1 = VL × IL = 2253 V × 0.2714 A = 611.5 W **Step 7**: Power2 = Vth × Ith = 4506 V × 0.2714 A = 1223 W **Step 8**: Maximum Power Calculation Pmax = $(Vth^2) / (4 \times Rth) = (4506\ V)^2 / (4 \times 8300\ \Omega) = 20304036 / 33200 = 611.57$ W **Step 10:** Verify Power Calculation Power1 = 611.5 W ≈ Pmax = 611.57 W ✓

Maximum Power Transfer Analysis - High Voltage Circuits

Given Thevenin Equivalent:

- Rth = 8.3 kΩ = 8300 Ω
- Vth = 4.506 kV = 4506 V

Maximum Power Transfer Condition:

RL = Rth = 8.3 kΩ

Circuit 1: RL = 6.8 kΩ **Calculations:** • Rtotal = Rth + RL = 8300 + 6800 = 15100 Ω • IL = Vth/Rtotal = 4506/15100 = 0.2984 A • VL = IL × RL = 0.2984 × 6800 = 2029.12 V • VRth = IL × Rth = 0.2984 × 8300 = 2476.88 V • Verification: Vth = VRth + VL = 2476.88 + 2029.12 = 4506 V ✓ • **PL = VL × IL = 2029.12 × 0.2984 = 605.5 W** • Psource = Vth × IL = 4506 × 0.2984 = 1344.6 W **Efficiency:** η = PL/Psource = 605.5/1344.6 = **45.0%**	**Circuit 2: RL = 3.3 kΩ** **Calculations:** • Rtotal = Rth + RL = 8300 + 3300 = 11600 Ω • IL = Vth/Rtotal = 4506/11600 = 0.3884 A • VL = IL × RL = 0.3884 × 3300 = 1281.72 V • VRth = IL × Rth = 0.3884 × 8300 = 3224.28 V • Verification: Vth = VRth + VL = 3224.28 + 1281.72 = 4506 V ✓ • **PL = VL × IL = 1281.72 × 0.3884 = 497.82 W** • Psource = Vth × IL = 4506 × 0.3884 = 1750.13 W **Efficiency:** η = PL/Psource = 497.82/1750.13 = **28.4%**
Circuit 3: RL = 1.1 kΩ **Calculations:** • Rtotal = Rth + RL = 8300 + 1100 = 9400 Ω • IL = Vth/Rtotal = 4506/9400 = 0.4793 A • VL = IL × RL = 0.4793 × 1100 = 527.23 V • VRth = IL × Rth = 0.4793 × 8300 = 3978.77 V • Verification: Vth = VRth + VL = 3978.77 + 527.23 = 4506 V ✓ • **PL = VL × IL = 527.23 × 0.4793 = 252.7 W** • Psource = Vth × IL = 4506 × 0.4793 = 2159.7 W **Efficiency:** η = PL/Psource = 252.7/2159.7 = **11.7%**	**Circuit 4: RL = 8.3 kΩ (MAXIMUM POWER CONDITION)** **Calculations:** • Rtotal = Rth + RL = 8300 + 8300 = 16600 Ω • IL = Vth/Rtotal = 4506/16600 = 0.2714 A • VL = IL × RL = 0.2714 × 8300 = 2253 V • VRth = IL × Rth = 0.2714 × 8300 = 2253 V • Verification: Vth = VRth + VL = 2253 + 2253 = 4506 V ✓ • **PL = VL × IL = 2253 × 0.2714 = 611.5 W** • Psource = Vth × IL = 4506 × 0.2714 = 1223 W **Efficiency:** η = PL/Psource = 611.5/1223 = **50.0%**

Maximum Power Transfer Verification:

Theoretical Maximum Power: $P_{L,max} = V_{th}^2/(4 \times R_{th}) = (4506)^2/(4 \times 8300) = 20304036/33200 =$ **611.57 W**

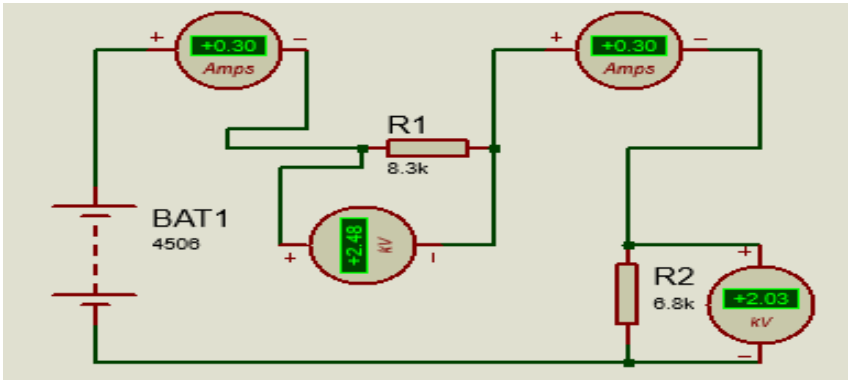
Verification: $P_L = 611.5$ W ≈ $P_{L,max} = 611.57$ W ✓

Summary Table:

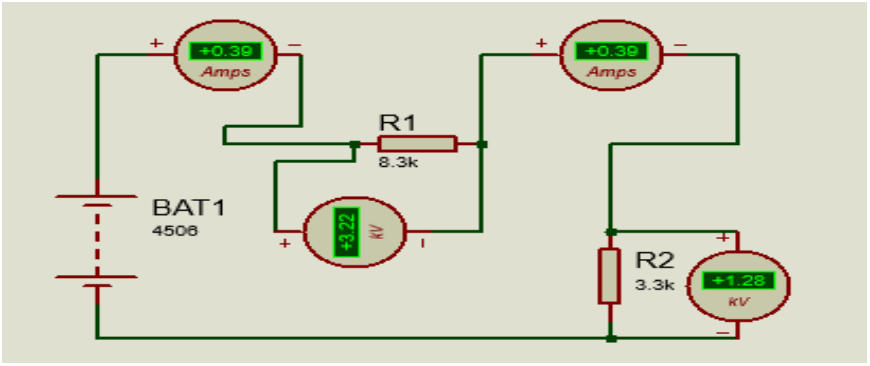
Circuit	RL (kΩ)	Current (A)	VL (V)	Power (W)	Efficiency (%)
1	6.8	0.2984	2029.12	605.5	45.0
2	3.3	0.3884	1281.72	497.82	28.4
3	1.1	0.4793	527.23	252.7	11.7
4	**8.3**	**0.2714**	**2253**	**611.5**	**50.0**

Protius Simulation

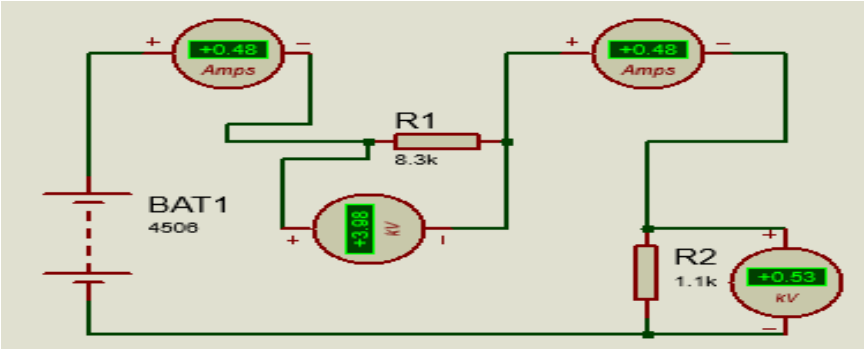
Thevenin Circuit 1: (6.8KΩ)



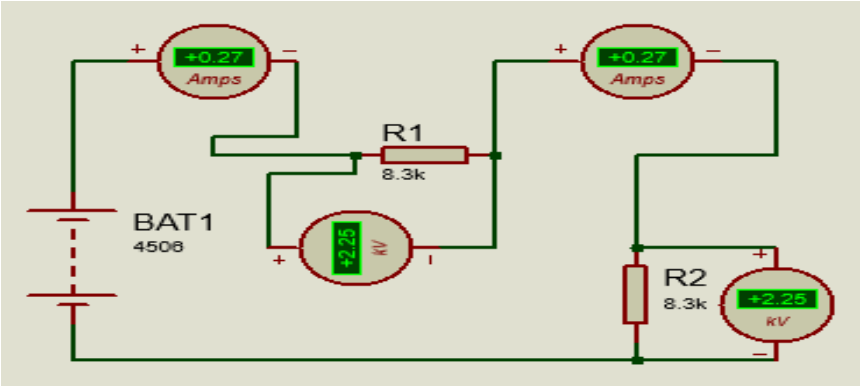
Thevenin Circuit 2(3.3KΩ)



Thevenin Circuit 3: (1.1KΩ)



Thevenin Circuit 4: (2.8KΩ)



Conclusion:

After performing the tasks and studying the Maximum Power Transfer Theorem, we can conclude that this rule helps us find the best condition to transfer the most power from a source to a load. It tells us that the load will receive maximum power when its resistance is equal to the source's internal resistance (called Thévenin or Norton resistance). This is useful in designing circuits like amplifiers or radios where getting the most output is important. However, when power transfer is at its maximum, the efficiency is only 50%, meaning half the power is lost in the source. So, in real-life applications, we must decide whether we want more power or more efficiency depending on the situation.